

# How to Determine your IR Controller Hex Codes

Saturday, September 5, 2026 12:10 PM

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If you are new to working with IR controllers and IR receivers, you may not know which hexadecimal (HEX) codes need to be entered into your Arduino sketch for your particular IR controller. The codes are specific to the controller, so they must be identified before the Arduino can be programmed to recognize the individual buttons.

To determine the codes, first upload the IR receiver test sketch to the Arduino and open the **Serial Monitor** in the Arduino IDE. Point the IR controller toward the IR receiver and press each button, one at a time. The Serial Monitor will display the HEX code received from the controller for each button.

Record each HEX code and the button that generated it. For example:

- Zone 1 button → 0XXXXXXXXX
- Zone 2 button → 0XXXXXXXXX
- Zone 3 button → 0XXXXXXXXX

Continue this process for every button you intend to use. It is a good idea to press each button several times to verify that the same code is consistently received.

Once all of the required codes have been identified, replace the example HEX codes in the Arduino sketch with the codes from your particular controller. The Arduino can then compare each received IR code with the corresponding value in the sketch and perform the appropriate function when that button is pressed.

**Note:** If you hold a button down, you may see a repeat code rather than the button's original HEX code. This is normal behavior with many IR protocols. For identifying the codes, use short, individual button presses.

The video linked below provides a visual walkthrough of this process, including how to use the Serial Monitor to identify the HEX codes transmitted by each button on an IR controller.

[Decode Hexadecimal Codes of IR Remote with Arduino | #arduino](#)

